ACADEMIC CENTRAL

PRACTICE EXAMINATION

UNIVERSITY · 3RD YEAR

MICROCONTROLLERS AND EMBEDDED SYSTEMS

PAPER 1 · PRACTICE A

MARKS: 100 **TIME: 3 hours**

BEng Mechatronics and Robotics Engineering, 3rd year final examination: microcontroller architecture and memory, address decoding, embedded C and bit manipulation, interrupts, timers and PWM, ADC and interfacing, serial communication (UART, SPI, I2C).

Written by Academic Central for practice, following embedded systems modules at South African universities (e.g. University of Pretoria EMK 310 Microprocessors: architecture, memory and address decoding, embedded programming, I/O interfacing, timers, ADCs, serial ports and PWM). It is not a university examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of FIVE questions. Answer ALL the questions.
2. Unless stated otherwise, assume an 8-bit AVR microcontroller (for example the ATmega328P on an Arduino Uno) with a 16 MHz clock and a 5 V supply. Register names follow the AVR datasheet.
3. Write code in C. Small syntax mistakes that do not change the logic will not be penalised.
4. Non-programmable calculators may be used. Show ALL your working.
5. Number your answers exactly as the questions are numbered.

QUESTION 1: ARCHITECTURE AND MEMORY

1.1 Explain the difference between a Harvard and a von Neumann architecture, and between a microcontroller and a microprocessor. (4)

1.2 Name THREE types of memory found on a microcontroller and say what each one is used for. (3)

1.3 A system has a 16-bit address bus. A peripheral is mapped to the addresses 0x8000 to 0x8FFF.

1.3.1 How many bytes does the peripheral occupy? (2)

1.3.2 Which address lines, and at what levels, must the decoder use to select it? (2)

1.3.3 How many address lines are needed to address a 32 KB memory chip? (1)

1.4 Write ONE C statement for each of the following, without changing any other bits: (a) set bit 5 of PORTB, (b) clear bit 2 of PORTB, (c) toggle bit 0 of PORTB. Then (d) write an if condition that is true when bit 3 of PINB is 1. (3)

[15]

QUESTION 2: EMBEDDED C, INPUTS AND INTERRUPTS

2.1 Why must a global variable that is changed inside an interrupt service routine (ISR) be declared volatile? (2)

2.2 Compare polling and interrupts for reading an input, and give one situation in a robot where each is the better choice. (3)

2.3 What is switch bounce, and how can it be handled in software? (3)

2.4 Give the value of a and of b after the code below runs. Explain each answer. (4)

```
uint8_t a = 200;
a = a + 100;
int8_t b = 127;
b++;
```

2.5 A push button on PD2 connects the pin to 0 V when pressed (use the internal pull-up resistor). An LED is connected to PB5. Write a complete program that toggles the LED once each time the button is pressed, with a simple software debounce. (6)

2.6 What is the value of 0b10110010 >> 3, in binary and in decimal? (2)

[20]

QUESTION 3: TIMERS AND PWM

3.1 A 16-bit timer runs from the 16 MHz clock with a prescaler of 64.

3.1.1 Calculate the time for one timer tick. (2)

3.1.2 Calculate the compare value (OCR1A) for an interrupt every 10 ms in CTC mode. (3)

3.2 An 8-bit timer runs with a prescaler of 256. Calculate the time between overflows. (3)

3.3 An 8-bit timer is used in fast PWM mode with a prescaler of 8.

3.3.1 Calculate the PWM frequency. (3)

3.3.2 Calculate the duty cycle when OCR0A = 191, and the average output voltage. (3)

3.4 A hobby servo needs a 50 Hz signal with a pulse width from 1 ms (0°) to 2 ms (180°). A 16-bit timer with a prescaler of 8 is used.

3.4.1 Calculate the pulse width for 45°. (2)

3.4.2 Calculate the TOP value (ICR1) for a 20 ms period. (2)

3.4.3 Calculate the compare value for the 45° pulse. (2)

3.5 An ultrasonic sensor's echo pulse is measured with the timer's input-capture unit and is 1 160 µs long. Explain how input capture measures the pulse, and calculate the distance to the obstacle (speed of sound 343 m/s). (5)

[25]

QUESTION 4: ADC AND INTERFACING

4.1 An LM35 temperature sensor gives 10 mV/°C and is read by the 10-bit ADC with a 5 V reference. The ADC result is 154.

4.1.1 Calculate the temperature. (3)

4.1.2 Calculate the smallest temperature change the system can detect. (2)

4.2 The internal 1.1 V reference is selected instead. Calculate the new resolution in °C and the highest temperature that can be measured. (3)

4.3 Describe how a successive-approximation ADC converts an input voltage. (3)

4.4 Explain why a 12 V DC motor may not be connected straight to a microcontroller pin, and describe a suitable driver circuit. (4)

4.5 A sensor with a 5 V output must be connected to a 3.3 V microcontroller input. Design a voltage divider with a 1.8 kΩ upper resistor, and calculate the voltage at the input when the sensor outputs 5 V. (3)

4.6 What is the purpose of a watchdog timer? (2)

[20]

QUESTION 5: SERIAL COMMUNICATION

5.1 A UART is set to 9 600 baud, 8 data bits, no parity, 1 stop bit (8N1).

5.1.1	How many bits are sent for each byte?	(1)
5.1.2	Calculate the maximum number of bytes per second.	(2)
5.1.3	Calculate the time to send the 7-character message "HELLO\r\n".	(2)
5.2	The baud-rate register is set with UBRR = f_clk / (16 × baud) − 1. Calculate UBRR for 9 600 baud, the actual baud rate obtained, and the percentage error.	(4)
5.3	Compare UART, SPI and I2C under the headings: number of wires, clock, how a device is selected, and typical speed.	(6)
5.4	An MPU-6050 IMU has the 7-bit I2C address 0x68. Give the address byte the master sends to write to it and to read from it, and explain why the I2C lines need pull-up resistors.	(3)
5.5	What is the purpose of the chip-select (CS) line on an SPI bus?	(2)

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TOTAL: 100

END OF QUESTION PAPER